



## AQ7.1: Activity Questions 1 - Not Graded



This assignment will not be graded and is only for practice.

### Level 1

1 point

Choose the set of correct options.

☐

If a  $3 \times 3 \times 3$  matrix  $AA$  is similar to the identity matrix of order 33, then  $AA$  must be the identity matrix of order 3.

☐

If a  $3 \times 3 \times 3$  matrix  $AA$  is similar to a diagonal matrix of order 33, then  $AA$  must be a diagonal matrix of order 3.

☐

If  $AA$  and  $BB$  are similar matrices, then they are also equivalent matrices.

☐

If  $AA$  and  $BB$  are equivalent matrices, then they are also similar matrices.

**No, the answer is incorrect.**

**Score: 0**

### Accepted Answers:

If a  $3 \times 3 \times 3$  matrix  $AA$  is similar to the identity matrix of order 33, then  $AA$  must be the identity matrix of order 3.

If  $AA$  and  $BB$  are similar matrices, then they are also equivalent matrices.

Consider the linear transformation given below:

$$T: R^3 \rightarrow R^2 \quad T: R^3 \rightarrow R^2$$

$$T(x, y, z) = (x - y, y + z) \quad T(x, y, z) = (x - y, y + z)$$

Consider two ordered bases  $\beta_1 = \{(1, 0, 0), (0, 1, 0), (0, 0, 1)\}$  and  $\beta_2 = \{(1, 0, 1), (0, 1, 1), (1, 1, 0)\}$  of  $R^3$  and consider the standard ordered basis  $\gamma = \{(1, 0), (0, 1)\}$  of  $R^2$ . Let  $AA$  and  $BB$  be the matrices corresponding to the linear transformation  $T$  with respect to the bases  $\beta_1$  and  $\beta_2$  for the domain, respectively and  $\gamma$  for the co-domain. Let  $QQ$  and  $PP$  be matrices such that  $B = QAPB = QAP$ .

Answer the questions 2,3, and 4 using the above information.

1 point

Which of the following matrices represents  $AA$ ?

☐

$$\begin{bmatrix} 1 & -1 & 0 \\ 1 & 2 & 1 \end{bmatrix}$$

☐

$$\begin{bmatrix} 1 & -1 & 0 \\ 0 & 1 & 1 \end{bmatrix}$$

☐

$$\begin{bmatrix} 1 & 1 \\ -1 & 1 \\ 0 & 1 \end{bmatrix}$$

☐


$$\begin{bmatrix} 1 & 1 \\ -1 & 2 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & -1 & 0 & 1 & 2 & 1 \end{bmatrix}$$

**No, the answer is incorrect.**

**Score: 0**

**Accepted Answers:**

$$\begin{bmatrix} 1 & -1 & 0 \\ 0 & 1 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & -1 & 1 & 0 & 1 \end{bmatrix}$$

1 point

Which of the following matrices can be  $PP$ ?

☐

There does not exist any matrix  $PP$  which satisfies the property  $B = QAPB = QAP$ .

☐

$$\begin{bmatrix} 1 & 0 & 1 \\ 0 & 0 & 1 \\ 1 & 2 & 0 \end{bmatrix} \begin{bmatrix} 1 & 0 & 1 & 0 & 0 & 2 & 1 & 1 & 0 \end{bmatrix}$$

☐

$$\begin{bmatrix} \frac{1}{2} & -\frac{1}{2} & \frac{1}{2} \\ -\frac{1}{2} & \frac{1}{2} & \frac{1}{2} \\ \frac{1}{2} & \frac{1}{2} & -\frac{1}{2} \end{bmatrix} \begin{bmatrix} 2 & 1 & -2 & 1 & 2 & 1 & -2 & 1 & 2 & 1 & 2 & 1 & -2 & 1 \end{bmatrix}$$

☐

$$\begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \\ 1 & 1 & 0 \end{bmatrix} \begin{bmatrix} 1 & 0 & 1 & 0 & 1 & 1 & 1 & 1 & 0 \end{bmatrix}$$

**No, the answer is incorrect.**

**Score: 0**

**Accepted Answers:**

$$\begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \\ 1 & 1 & 0 \end{bmatrix} \begin{bmatrix} 1 & 0 & 1 & 0 & 1 & 1 & 1 & 1 & 0 \end{bmatrix}$$

1 point

Which of the following statements is true for  $QQ$ ?

☐

$QQ$  can be the identity matrix of order 3.

☐

$QQ$  can be the identity matrix of order 2.

☐

Whatever the matrix  $PP$  we choose,  $QQ$  is always unique.

☐

$QQ$  can be the matrix  $\begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} 0 & 1 & 1 & 0 \end{bmatrix}$ .

**No, the answer is incorrect.**

**Score: 0**

**Accepted Answers:**

QQ can be the identity matrix of order 2.

QQ can be the matrix  $\begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$ .

1 point

Choose the set of correct options.

☐ If two square matrices of the same order have the same determinants, then they must be similar to each other.

☐ Similar matrices have the same rank.

☐

If  $AA$  and  $BB$  are similar matrices, and  $Ax = bAx = b$  has a unique solution, then  $Bx = bBx = b$  also has a unique solution.

**No, the answer is incorrect.**

**Score: 0**

**Accepted Answers:**

Similar matrices have the same rank.

If  $AA$  and  $BB$  are similar matrices, and  $Ax = bAx = b$  has a unique solution, then  $Bx = bBx = b$  also has a unique solution.

1 point

Choose the set of correct options.

☐

If  $\det AA$  or  $\det BB$  is nonzero, then  $ABAB$  and  $BABA$  are similar.

☐

$A = \begin{bmatrix} 2 & 1 \\ 5 & 3 \end{bmatrix}$   $A = [2513]$  and  $B = \begin{bmatrix} 2 & 3 \\ 1 & 2 \end{bmatrix}$   $B = [2132]$  are similar.

☐

$A = \begin{bmatrix} 2 & 1 \\ 0 & 3 \end{bmatrix}$   $A = [2013]$  and  $B = \begin{bmatrix} 1 & 1 & 1 \\ 0 & 4 \end{bmatrix}$   $B = [10114]$  are similar.

☐

$A = \begin{bmatrix} 2 & 1 \\ 0 & 2 \end{bmatrix}$   $A = [2012]$  and  $B = \begin{bmatrix} 2 & 0 \\ 0 & 2 \end{bmatrix}$   $B = [2002]$  are not similar.

**No, the answer is incorrect.**

**Score: 0**

**Accepted Answers:**

If  $\det AA$  or  $\det BB$  is nonzero, then  $ABAB$  and  $BABA$  are similar.

$A = \begin{bmatrix} 2 & 1 \\ 0 & 2 \end{bmatrix}$   $A = [2012]$  and  $B = \begin{bmatrix} 2 & 0 \\ 0 & 2 \end{bmatrix}$   $B = [2002]$  are not similar.

1 point

Which of the following options are correct?

☐ Two invertible matrices of same order are equivalent.

☐

$A = \begin{bmatrix} 17 & 6 \\ 21 & -1 \\ 52 & -3 \end{bmatrix}$   $A = [1257126-1-3]$  and  $B = \begin{bmatrix} 23 & -2 \\ 11 & 4 \\ 12 & -6 \end{bmatrix}$   $B = [211312-24-6]$  are equivalent.

☐

$A = \begin{bmatrix} 2 & 3 \\ 4 & 6 \end{bmatrix}$   $A = [2436]$  and  $B = \begin{bmatrix} 1 & 3 \\ 3 & 1 \end{bmatrix}$   $B = [1331]$  are equivalent.

☐

Consider two linear transformations  $T_1(x, y, z) = (x + y - z, x + 4y - 2z, -2x + y + z)$  and  $T_2(x, y, z) = (x + y, x - z, y + z)$ . Let  $A$  and  $B$  denote the matrix representation of  $T_1$  and  $T_2$  with respect to the standard basis of  $\mathbb{R}^3$ . Then  $A$  and  $B$  are equivalent.

**No, the answer is incorrect.**

**Score: 0**

**Accepted Answers:**

Two invertible matrices of same order are equivalent.

$$A = \begin{bmatrix} 1 & 1 & -1 \\ 1 & 4 & -2 \\ -2 & 1 & 1 \end{bmatrix} \text{ and } B = \begin{bmatrix} 1 & 1 & 0 \\ 1 & 0 & -1 \\ 0 & 1 & 1 \end{bmatrix} \text{ are equivalent.}$$

Consider two linear transformations  $T_1(x, y, z) = (x + y - z, x + 4y - 2z, -2x + y + z)$  and  $T_2(x, y, z) = (x + y, x - z, y + z)$ . Let  $A$  and  $B$  denote the matrix representation of  $T_1$  and  $T_2$  with respect to the standard basis of  $\mathbb{R}^3$ . Then  $A$  and  $B$  are equivalent.

## Level 2

1 point

Let  $E$  be the set of  $2 \times 2$  matrices which are similar to a scalar matrix  $A = \begin{bmatrix} 3 & 0 \\ 0 & 3 \end{bmatrix}$ .

Choose the set of correct statements.

☐

$E$  is an infinite set.

☐

Cardinality of  $E$  is finite.

☐

$E$  is singleton set.

☐

$E$  consists of all the scalar matrices of order  $n$ .

**No, the answer is incorrect.**

**Score: 0**

**Accepted Answers:**

Cardinality of  $E$  is finite.

$E$  is singleton set.

1 point

Which of the following option(s) is(are) true?

☐

If  $A$  and  $B$  are similar matrices then  $A^{-1}$  and  $B^{-1}$  are similar matrices.

☐

If  $A$  and  $B$  are similar matrices then  $A^2$  and  $B^2$  are similar matrices.

☐

If  $A^2$  and  $B^2$  are similar matrices then  $A$  and  $B$  are similar matrices.

☐

If  $A$  and  $B$  are similar matrices then  $A^T$  and  $B^T$  are similar matrices.

**No, the answer is incorrect.**

**Score: 0**

**Accepted Answers:**

If  $A$  and  $B$  are similar matrices then  $A^{-1}$  and  $B^{-1}$  are similar matrices.

If  $A$  and  $B$  are similar matrices then  $A^2$  and  $B^2$  are similar matrices.

If  $A$  and  $B$  are similar matrices then  $A^T$  and  $B^T$  are similar matrices.

1 point

Let us consider the following matrices.

$$A = \begin{bmatrix} 1 & 0 \\ 1 & 1 \end{bmatrix}, B = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}, C = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix} \quad A = [1101], B = [1001], C = [1011]$$

Choose the set of correct options.

☐

$AA$  and  $BB$  have the same rank, and same determinant, but they are not similar matrices.

☐

$AA$  and  $CC$  have the same rank, and same determinant, but they are not similar matrices.

☐

$BB$  and  $CC$  have the same rank, and same determinant, but they are not similar matrices.

☐ None of the options are true.

**No, the answer is incorrect.**

**Score: 0**

**Accepted Answers:**

$AA$  and  $BB$  have the same rank, and same determinant, but they are not similar matrices.

$BB$  and  $CC$  have the same rank, and same determinant, but they are not similar matrices.

[Check Answers](#)

Your score is: 0/10